CSE 4137

Machine Learning
Module-01
Multiple Regression



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Topics: Supervised ML

Week-03

- Multiple Linear Regression
- Example

Multiple Regression: SML

Week-04, c-1

Multiple Linear Regression

What Is (MLR)?

Multiple linear regression (MLR), also known simply as multiple regression, is a statistical technique that uses several explanatory variables to predict the outcome of a response variable. The goal of multiple linear regression is to model the linear relationship between the explanatory (independent) variables and response (dependent) variables

Formula and Calculation of Multiple Linear Regression

$$y_i = \beta_0 + \beta_1 x_{i1} + \beta_2 x_{i2} + ... + \beta_p x_{ip} + \epsilon$$

where, for $i = n$ observations:

y_i = dependent variable

x_i = explanatory variables

β_0 = y-intercept (constant term)

β_p = slope coefficients for each explanatory variable

ϵ = the model's error term (also known as the residuals)

The Difference Between Linear and Multiple Regression

Regression compares the response of a dependent variable given a change in some explanatory variables. However, a dependent variable is rarely explained by only one variable. In this case, an analyst uses multiple regression, which attempts to explain a dependent variable using more than one independent variable. Multiple regressions can be linear and nonlinear.

Why Would One Use a Multiple Regression Over a Simple OLS Regression?

A dependent variable is rarely explained by only one variable. In such cases, an analyst uses multiple regression, which attempts to explain a dependent variable using more than one independent variable. The model, however, assumes that there are no major correlations between the independent variables.

Multiple Regression: SML

Week-04, C-1

Example

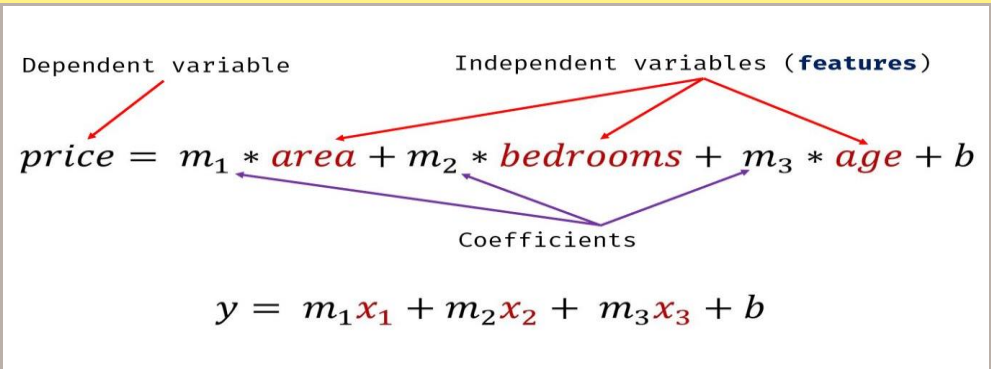
The problem is to predict the price of a house using a given dataset. Here **price** depends on **area** (square feet), **bed rooms** and **age** of the home (in years). Given these prices we have to predict prices of new homes based on area, bed rooms and age.

Given these home prices find out price of a home that has,

3000 sqr ft area, 3 bedrooms, 40 year old

2500 sqr ft area, 4 bedrooms, 5 year old

Price can be calculated using following equation,



Import libraries

```
import pandas as pd
import numpy as np
from sklearn import linear_model
```

Load Dataset

```
df = pd.read_csv('homeprices.csv')
df
```

	area	berooms	age	price
0	2600	3.0	20	550000
1	3000	4.0	15	565000
2	3200	NaN	18	610000
3	3600	3.0	30	696000
4	4000	5.0	8	760000

Multiple Regression: SML

Week-04, c-2

Multivariate Regression

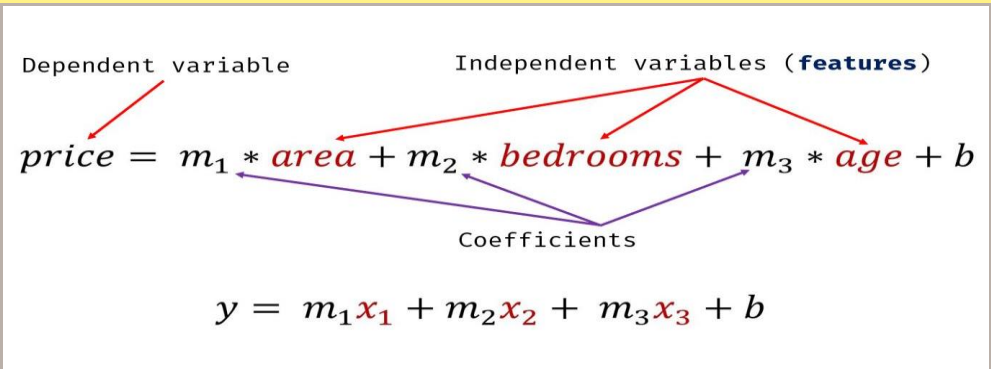
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Multiple Regression: SML

Week-04, c-2

Handle Missing Data

```
df.bedrooms.median()
df.bedrooms = df.bedrooms.fillna(df.bedrooms.median())
df
```

	area	bedrooms	age	price
0	2600	3.0	20	550000
1	3000	4.0	15	565000
2	3200	4.0	18	610000
3	3600	3.0	30	696000
4	4000	5.0	8	760000
5	4100	6.0	8	810000

Train Model

```
inputs=df.drop('price',axis='columns')
price=df.price
reg = linear_model.LinearRegression()
reg.fit(inputs,price)
```

Find coef_ and intercept_

```
reg.coef_
reg.intercept_
```

Predict Price

```
#Find price of home with 3000 sqr ft area, 3
# bedrooms, 40 year old
reg.predict([[3000, 3, 40]])
```

Check Price

```
12.06244194*3000 + 23388.88007794*3 +
3231.71790863*40 + 221323.00186540384
```

Check Metrics

```
y_pred=reg.predict(inputs)
from sklearn.metrics import
mean_absolute_error,mean_squared_error,r2_score
print('MAE=',mean_absolute_error(price,y_pred))
print('MSE=',mean_squared_error(price,y_pred))
print('RMSE=',np.sqrt(mean_squared_error(price,y_pred)))
print("R2_Score=",r2_score(price,y_pred))
```

Home Work: SML

Week-04, c-2

Predict the Salary

An IT firm keeps record of its employees such as experience of candidate, his written test score and personal interview score. Based on these 3 factors, HR will decide the salary. Given this data, you need to build a machine learning model for HR department that can help them decide salaries for future candidates. Using this predict salaries for following candidates,

2 yr experience, 9 test score, 6 interview score

12 yr experience, 10 test score, 10 interview score

Hint: Create a .CSV file with the features given and use python program to predict the salary

End of Chapter